Data and network construction

- DisGeNET: curated gene-disease database
- C0014175 = UMLS endometriosis ID
- STRING: physical and functional protein interactions
- First-shell expansion: adds direct neighbours, cap 300
- Density driven by the expansion

Null models

- ER, Erdős-Rényi: random edges at endo-ppi's density; tests "edge count alone?"
- BA, Barabási-Albert: preferential attachment with $m \approx 53$, -> so total edge count matches PPI; tests "scale-free distribution alone?"
- Pref attachment: rich-get-richer; each new node adds m edges to existing nodes with probability $\propto$ degree, producing a scale-free distribution
- Scale-free: power-law degree distribution $P(k) \propto k^{-\alpha}$; many low-degree nodes plus a few high-degree hubs, no typical degree
- CM, configuration model: preserves degree sequence; tests "degrees alone?"
- CM by double-edge swap: pick two edges, swap their endpoints; preserves every node's degree while randomizing topology
- `nx.configuration_model` would lose 20% of edges in dense graph, due to parallel edges and self-loops collapsing on simplification

Network statistics

- Clustering: fraction of a node's neighbor pairs that are also connected
- PPI 0.637 vs CM 0.512, same degree sequence; 25% gap, so ppi not so random
- Assortativity: degree correlation at edge endpoints
- PPI uniquely positive, +0.052; hubs cluster together
- CM disassortative, −0.181; too few hubs to pair with each other randomly

Degree distribution

- PPI heavy-tailed: low-degree nodes + extreme hubs, highest = 100 * lowest -> span two orders of magnitude
- ER: narrow Poisson peak around mean 93, no tail
- BA: power-law shape by design but scattered here, minimum degree fixed at $m \approx 53$
- CM matches PPI by construction

Diffusion models

- SIR: each step infected node infect each neighbor with prob β, then recovers with prob γ

- IC: newly activated node gets one chance per inactive neighbor with prob p, then exhausted
- Both are Monte Carlo simulations, 50 runs per config, results averaged
- Pairing the two checks whether results depend on choice of dynamics

Calibration sweep

- Too high: network saturates, every blocking strat looks ineffective
- Too low: propagation dies, every strat looks artificially good
- Target: 60–85% of the network without blocking
- Dense graph forces pretty low parameters.
- Each row = mean over 50 Monte Carlo runs at fixed parameters

Group-centrality algorithms

- Structural: look only at graph topology, no diffusion simulation
- "Lazy" in CELF = recompute spread only for candidates that could be stale; cuts the cost by orders of magnitude
- the problem of finding truly optimal set is NP-hard
- gdeg attacks a submodular function, so greedy achieves at least $(1-1/e)$ of the optimum (Nemhauser et al. 1978)
- for each k total 4 result chains: gdeg, gbtw, CELF + SIR, CELF + IC

Top 10 picks per chain

- gdeg & gbtw share IL6, ESR1, EGFR, INS in top-10: topological quality
- ESR1 is the estrogen receptor, biologically central to endometriosis
- `celf_sir` top picks are growth-factor / angiogenesis proteins (FGF4, KDR, PDGFRB)
- `celf_ic` top picks are immune-modulation proteins (CD40, LAG3, CX3CL1)
- CELF chains share 0 with each other and 0 with structural chains in top-10
- gdeg-gbtw overlap of 4/10 is well above chance; everything else at chance baseline

Spread reduction across algorithms

- gbtw plateau at $k \approx 200$, hits 98.7% SIR / 99.1% IC
- Next-best (celf_ic) behind by 12–17 pp at $k = 200$
- Random baseline lowest at every $k \geq 100$
- CELF underperforms despite optimizing the spread metric directly
- Why CELF loses: dense PPI rewards high-betweenness hubs

PPI vs null networks

- PPI vs CM (same degree sequence) under gbtw: 7.8 pp gap = real higher-order structure
- BA vulnerable to gdeg because extreme hubs fragment the graph fast

- PPI doesn't show that vulnerability despite heavy tail; clustering buffers it

Top-20 composition

- Union of top-20 across 4 chains: 75 distinct proteins
- Only 5 appear in 2+ chains, all gdeg-gbtw: ESR1, IL6, EGFR, INS, CYP3A4
- Every other pairwise comparison: zero overlap in top-20
- 93% algorithm-specific = answer is dominated by which algorithm you ask
- celf_sir cluster: angiogenesis + ECM remodeling (FGF4, KDR, PDGFRB, VTN, TIMP1)
- celf_ic cluster: immune modulation + transcription (CD40, LAG3, HDAC2, MED4)

Conclusion and limitations

- Best k-set: prefix of gbtw chain, saturates around $k = 200$ (44% of nodes)
- Strongly algorithm-dependent: 93% of top-20 algorithm-specific, CELF chains share 0 with structural
- At realistic small k (10 proteins), even best algorithm achieves only 7% reduction
- Diffusion-aware chains surface angiogenesis (SIR) and immune-modulation (IC) targets that structural methods miss